Ore is a naturally occuring rock found in Earth's crust that contains valuable minerals and metals. In an experiment, a researcher dissolved a sample of ore containing an unknown amount of iron in HCl(*aq*). The resulting aqueous solution, which contained both Fe^{2+}(*aq*) and Fe^{3+}(*aq*), was poured through a Jones reductor column containing Zn(Hg)(*s*) to convert all iron ions into a single oxidation state. The conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*) in the Jones reductor column is shown in Figure 1.

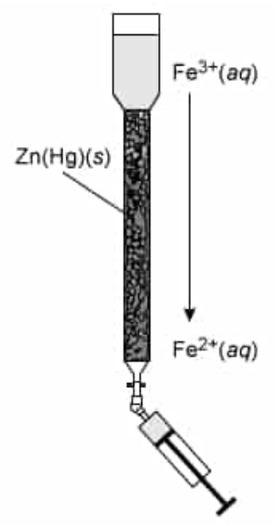

Figure 1 Jones reductor column used for the conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*). The resulting Fe^{2+}(*aq*) solution was collected using a syringe.

To determine the total amount of iron in the sample, the Fe^{2+}(*aq*) solution was then transferred to a container and titrated with 0.1 M $K_2Cr_2O_7$(*aq*), which resulted in the conversion of Fe^{2+}(*aq*) to Fe^{3+}(*aq*) in an irreversible process. A reference electrode was used to monitor the solution potential as a function of $K_2Cr_2O_7$(*aq*) volume added, and the equivalence point of the solution was reached after the addition of 9.50 mL of $K_2Cr_2O_7$(*aq*). A schematic of the experimental setup is shown in Figure 2.

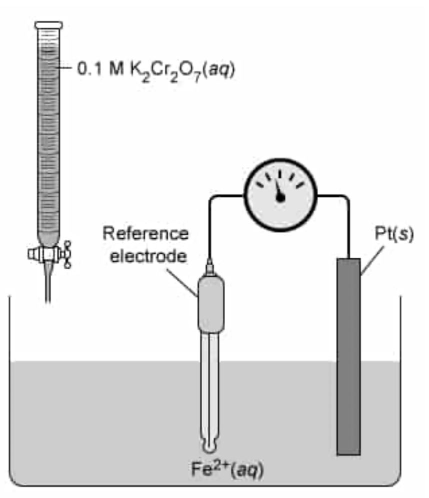

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Figure 2 Experimental setup for the titration of Fe^{2+}(*aq*) with $K_2Cr_2O_7$(*aq*) under acidic conditions

In a second experiment, the researcher prepared another sample of Fe^{2+}(*aq*) but used a 0.1 M I_3^-(*aq*) titrant solution instead of $K_2Cr_2O_7$(*aq*).

The standard reduction potentials for the species involved in the experiments are given in Table 1.

Table 1 Standard reduction potentials

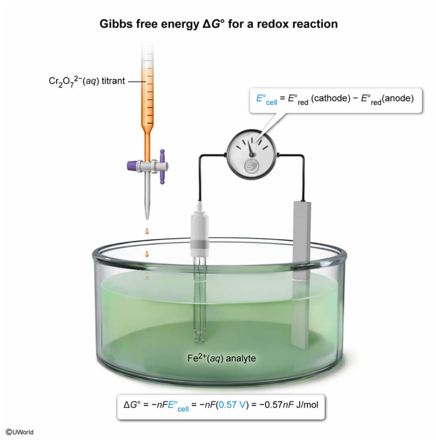
Reduction half-reaction	$E°$(V)
$Cr_2O_7^{2-}(aq) + 14\ H^+(aq) + 6\ e^- \rightarrow 2\ Cr^{3+}(aq) + 7\ H_2O(l)$	+1.33
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+0.76
$I_3^-(aq) + 2\ e^- \rightarrow 3\ I^-(aq)$	+0.53
$Zn^{2+}(aq) + Hg(l) + 2\ e^- \rightarrow Zn(Hg)(s)$	−0.76

Does the redox reaction between $Cr_2O_7^{2-}$(*aq*) and Fe^{2+}(*aq*) require energy input?

- A. Yes; $\Delta G°$ = +2.09*nF* J/mol. [17%]
- B. Yes; $\Delta G°$ = +0.57*nF* J/mol. [27%]
- ✔ C. No; $\Delta G°$ = −0.57*nF* J/mol. [37%]
- D. No; $\Delta G°$ = −2.09*nF* J/mol. [17%]

Correct
36% answered correctly

Explanation:



The change in the standard Gibbs free energy $\Delta G°$ associated with an oxidation–reduction (redox) reaction is related to the standard cell potential $E°_{cell}$ of the reaction by:

$$\Delta G° = -nFE°_{cell}$$

where *n* is the number of electrons transferred per reaction and *F* is Faraday's constant. A negative $\Delta G°$ (a **positive $E°_{cell}$**) indicates a thermodynamically favorable (ie, spontaneous) reaction, whereas a positive $\Delta G°$ (a **negative $E°_{cell}$**) indicates an unfavorable (ie, nonspontaneous) reaction in which external energy must be supplied to the system.

The standard cell potential $E°_{cell}$ is calculated by taking the difference between the standard **reduction potentials** of the reduction reaction $E°_{red}$(cathode) and the oxidation reaction $E°_{red}$(anode):

$$E°_{cell} = E°_{red}(\text{cathode}) - E°_{red}(\text{anode})$$

According to the passage, titrating Fe^{2+}(*aq*) with $Cr_2O_7^{2-}$(*aq*) produces Fe^{3+}(*aq*). As such, Fe^{2+}(*aq*) is oxidized (ie, electrons are lost) at the anode and $Cr_2O_7^{2-}$(*aq*) is reduced (ie, electrons are gained) at the cathode. Substituting the

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standard reduction potentials from Table 1 into the above equation yields:

$$E°_{cell} = 1.33\,V - 0.76\,V = 0.57\,V$$

Therefore, $\Delta G°$ for the redox reaction (expressed in terms of Faraday's constant and the number of electrons transferred) is:

$$\Delta G° = -nFE°_{cell}$$

$$\Delta G° = -\left(n\,\cancel{e^-}\right)\left(F\frac{J}{\cancel{V}\cdot mol\,\cancel{e^-}}\right)\left(+0.57\,\cancel{V}\right)$$

$$\Delta \boldsymbol{G}° = -0.57nF\frac{\mathbf{J}}{mol}$$

Based on this, the reaction is spontaneous, and no energy input is needed.

(Choices A and D) A value of 2.09nF J/mol is obtained if the reduction potentials are added rather than *subtracted* ($E°_{cell} = E°_{red}$(cathode) + $E°_{red}$(anode)).

(Choice B) If $E°_{cell}$ is incorrectly calculated as $E°_{red}$(anode) − $E°_{red}$(cathode), a positive $\Delta G°$ is obtained.

Educational objective:
The Gibbs free energy $\Delta G°$ of a reduction–oxidation reaction is related to the number of electrons n transferred in the overall reaction, the overall cell potential $E°_{cell}$, and Faraday's constant F, according to the equation $\Delta G° = -nFE°_{cell}$. For irreversible reactions, a positive $\Delta G°$ (negative $E°_{cell}$) indicates a nonspontaneous reaction whereas a negative $\Delta G°$ (positive $E°_{cell}$) indicates a spontaneous reaction.

Time spent:0 sec
Subject:
General Chemistry

QID:404098
System:
Solutions and Electrochemistry

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Ore is a naturally occuring rock found in Earth's crust that contains valuable minerals and metals. In an experiment, a researcher dissolved a sample of ore containing an unknown amount of iron in HCl(*aq*). The resulting aqueous solution, which contained both $Fe^{2+}(aq)$ and $Fe^{3+}(aq)$, was poured through a Jones reductor column containing Zn(Hg)(*s*) to convert all iron ions into a single oxidation state. The conversion of $Fe^{3+}(aq)$ to $Fe^{2+}(aq)$ in the Jones reductor column is shown in Figure 1.

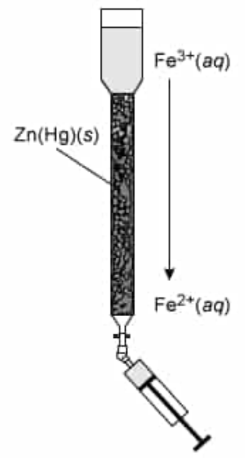

Figure 1 Jones reductor column used for the conversion of $Fe^{3+}(aq)$ to $Fe^{2+}(aq)$. The resulting $Fe^{2+}(aq)$ solution was collected using a syringe.

To determine the total amount of iron in the sample, the $Fe^{2+}(aq)$ solution was then transferred to a container and titrated with 0.1 M $K_2Cr_2O_7(aq)$, which resulted in the conversion of $Fe^{2+}(aq)$ to $Fe^{3+}(aq)$ in an irreversible process. A reference electrode was used to monitor the solution potential as a function of $K_2Cr_2O_7(aq)$ volume added, and the equivalence point of the solution was reached after the addition of 9.50 mL of $K_2Cr_2O_7(aq)$. A schematic of the experimental setup is shown in Figure 2.

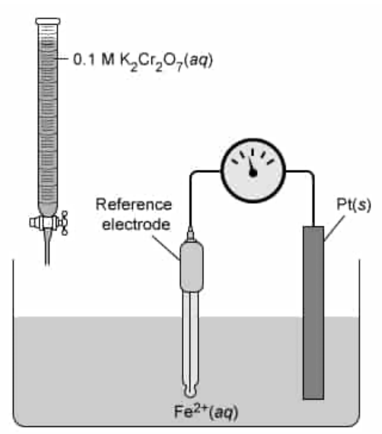

Figure 2 Experimental setup for the titration of $Fe^{2+}(aq)$ with $K_2Cr_2O_7(aq)$ under acidic conditions

In a second experiment, the researcher prepared another sample of $Fe^{2+}(aq)$ but used a 0.1 M $I_3^-(aq)$ titrant solution instead of $K_2Cr_2O_7(aq)$.

The standard reduction potentials for the species involved in the experiments are given in Table 1.

Table 1 Standard reduction potentials

Reduction half-reaction	$E°$(V)
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$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+0.76
$I_3^-(aq) + 2\ e^- \rightarrow 3\ I^-(aq)$	+0.53
$Zn^{2+}(aq) + Hg(l) + 2\ e^- \rightarrow Zn(Hg)(s)$	−0.76

The net reaction occurring in the Jones reductor column is:

- A. $Fe^{3+}(aq) + Zn(Hg)(s) + e^- \rightarrow Fe^{2+}(aq) + Zn^{2+}(aq) + Hg(l) + 2\ e^-$ [19%]
- B. $Fe^{3+}(aq) + Zn^{2+}(aq) + Hg(l) \rightarrow Fe^{2+}(aq) + Zn(Hg)(s)$ [5%]
- ✔ C. $2\ Fe^{3+}(aq) + Zn(Hg)(s) \rightarrow 2\ Fe^{2+}(aq) + Zn^{2+}(aq) + Hg(l)$ [60%]
- D. $2\ Fe^{3+}(aq) + Zn(Hg)(s) + 2\ e^- \rightarrow 2\ Fe^{2+}(aq) + Zn^{2+}(aq) + Hg(l) + 2\ e^-$ [14%]

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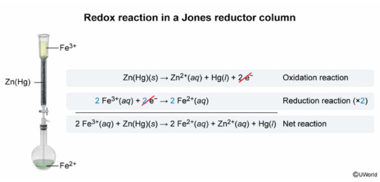
Correct
59% answered correctly

Explanation:



An oxidation–reduction (redox) reaction is a reaction in which one reactant is oxidized while another reactant is simultaneously **reduced**. As such, the **overall redox reaction** is the **sum of** the oxidation and reduction **half-reactions**. In a **balanced redox reaction**, the total number of **electrons released** from the oxidation half-reaction **must** be **equal** to the number of **electrons used** by the reduction half-reaction.

As shown in Figure 1, a Jones reductor column consists of a column containing Zn(Hg)(*s*) particles. When a solution containing Fe^{3+} ions is poured through the column, the Fe^{3+} ions are reduced from an oxidation state of +3 to +2 (ie, one electron gained per Fe^{3+})

Reduction half-reaction: $Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$

The electrons needed to reduce Fe^{3+} ions are generated through the oxidation of zinc atoms in Zn(Hg) (the reverse of the reduction reaction shown in Table 1). The zinc atoms are oxidized and go from an oxidation state of 0 to +2 (ie, two electrons lost per zinc atom).

Oxidation half-reaction: $Zn(Hg)(s) \rightarrow Zn^{2+}(aq) + Hg(l) + 2\ e^-$

The net reaction can be found by adding the two half-reactions together. To cancel the electron terms, the reduction half-reaction is multiplied by two (a multiple of both half-reactions), which yields a net reaction in which both **the atoms and the charges are balanced** (+6 charge on each side).

$$\begin{array}{rl}
2 \times \text{Reduction half-reaction:} & 2\ Fe^{3+}(aq) + \cancel{2\ e^-} \rightarrow 2\ Fe^{2+}(aq) \\
\text{Oxidation half-reaction:} & Zn(Hg)(s) \rightarrow Zn^{2+}(aq) + Hg(l) + \cancel{2\ e^-} \\
\hline
\text{Net:} & \mathbf{2\ Fe^{3+}(aq) + Zn(Hg)(s) \rightarrow 2\ Fe^{2+}(aq) + Zn^{2+}(aq) + Hg(l)}
\end{array}$$

(Choice A) In this reaction, the electrons are not balanced (1 e^- on the reactant side and 2 e^- on the product side).

(Choice B) The $Zn^{2+}(aq)$, Hg(*l*), and Zn(Hg)(*s*) terms are on wrong sides of the equation. This answer is obtained if the reduction half-reaction for Zn(Hg)(*s*) shown in Table 1 is not reversed.

(Choice D) Although the atoms and charges are balanced, the net reaction does not include the electron terms.

Educational objective:
An overall reduction–oxidation reaction can be expressed as the sum of the balanced oxidation and reduction half-reactions (or multiples of them).

Time spent: 0 sec
Subject:
General Chemistry

QID: 404099
System:
Solutions and Electrochemistry

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Ore is a naturally occuring rock found in Earth's crust that contains valuable minerals and metals. In an experiment, a researcher dissolved a sample of ore containing an unknown amount of iron in HCl(*aq*). The resulting aqueous solution, which contained both Fe^{2+}(*aq*) and Fe^{3+}(*aq*), was poured through a Jones reductor column containing Zn(Hg)(*s*) to convert all iron ions into a single oxidation state. The conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*) in the Jones reductor column is shown in Figure 1.

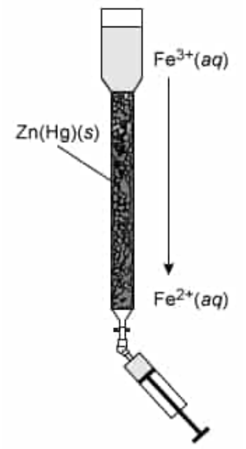

Figure 1 Jones reductor column used for the conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*). The resulting Fe^{2+}(*aq*) solution was collected using a syringe.

To determine the total amount of iron in the sample, the Fe^{2+}(*aq*) solution was then transferred to a container and titrated with 0.1 M $K_2Cr_2O_7$(*aq*), which resulted in the conversion of Fe^{2+}(*aq*) to Fe^{3+}(*aq*) in an irreversible process. A reference electrode was used to monitor the solution potential as a function of $K_2Cr_2O_7$(*aq*) volume added, and the equivalence point of the solution was reached after the addition of 9.50 mL of $K_2Cr_2O_7$(*aq*). A schematic of the experimental setup is shown in Figure 2.

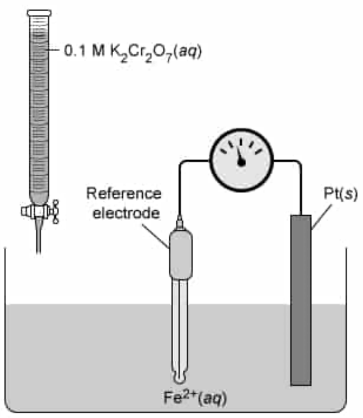

Figure 2 Experimental setup for the titration of Fe^{2+}(*aq*) with $K_2Cr_2O_7$(*aq*) under acidic conditions

In a second experiment, the researcher prepared another sample of Fe^{2+}(*aq*) but used a 0.1 M I_3^-(*aq*) titrant solution instead of $K_2Cr_2O_7$(*aq*).

The standard reduction potentials for the species involved in the experiments are given in Table 1.

Table 1 Standard reduction potentials

Reduction half-reaction	*E*°(V)
$Cr_2O_7^{2-}(aq) + 14\ H^+(aq) + 6\ e^- \rightarrow 2\ Cr^{3+}(aq) + 7\ H_2O(l)$	+1.33
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+0.76
$I_3^-(aq) + 2\ e^- \rightarrow 3\ I^-(aq)$	+0.53
$Zn^{2+}(aq) + Hg(l) + 2\ e^- \rightarrow Zn(Hg)(s)$	−0.76

What happens to the chromium atoms in $K_2Cr_2O_7$(*aq*) during the titration in the passage?

- A. Cr^{6+} is oxidized to Cr^{3+} [9%]
- B. Cr^{7+} is oxidized to Cr^{3+} [2%]
- ✔ C. Cr^{6+} is reduced to Cr^{3+} [77%]
- D. Cr^{7+} is reduced to Cr^{3+} [9%]

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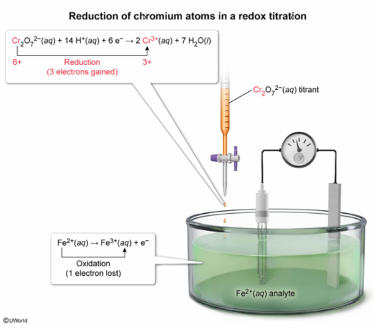
Correct
77% answered correctly

Explanation:



During an oxidation–reduction (redox) reaction, electrons are **transferred** from one atom to another. The atom that **loses** electrons is **oxidized** by the reaction and the atom that **gains** electrons is **reduced**. The oxidation state of the atom that becomes oxidized will **increase** by the number of electrons it loses and the oxidation state of the reduced atom will **decrease** by the number of electrons it gains. Oxidation states of the atoms in a compound can be determined using the compound formula and oxidation state rules. The **sum** of all the **oxidation states** of each atom in the compound must be equal to the **net charge** of the compound.

The passage states that titrating Fe^{2+}(*aq*) with $Cr_2O_7^{2-}$(*aq*) produces Fe^{3+}(*aq*). As such, Fe^{2+}(*aq*) becomes oxidized and therefore $Cr_2O_7^{2-}$(*aq*) becomes reduced. According to the reduction half-reaction for $Cr_2O_7^{2-}$, the final oxidation state of the Cr atom is +3 (ie, Cr^{3+}).

Applying the oxidation state rules to the $Cr_2O_7^{2-}$ anion shows that the initial oxidation state of the Cr atom is +6. Although Cr is a transition state metal with a variable oxidation state, the oxidation state of O (−2) is consistent and can be used to determine the oxidation state of the Cr atoms. Because $Cr_2O_7^{2-}$ has a net charge of −2, the oxidation states of Cr and O must sum to −2.

$$Cr_2O_7^{2-} = -2$$

$$2(Cr) + 7(O) = -2$$

$$2(Cr) + 7(-2) = -2$$

$$2(Cr) - 14 = -2$$

$$Cr = +6$$

Therefore, **$Cr^{6+}$is reduced to Cr^{3+}** (three electrons gained to form Cr^{3+}).

(Choice A) The oxidation state of chromium decreases from +6 to +3, which indicates that chromium *gains* 3 negatively charged electrons. Consequently, chromium is *reduced* during the titration.

(Choices B and D) If the oxidation state of Cr in $Cr_2O_7^{2-}$ was +7, the net charge of the compound would be 0 (ie, 2(+7) + 7(−2) = 0). However, the net charge of the anion is −2. Therefore, the initial oxidation state of Cr cannot be +7.

Educational objective:
The oxidation state of each atom in a compound or ion formula can be determined using oxidation state rules. The sum of the oxidation states of each atom must equal the net charge of the compound or ion. An increased oxidation state indicates a loss of electrons (oxidation) whereas a decreased oxidation state results from gaining electrons (reduction).

Time spent:0 sec
Subject:
General Chemistry

QID:404100
System:
Solutions and Electrochemistry

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Ore is a naturally occuring rock found in Earth's crust that contains valuable minerals and metals. In an experiment, a researcher dissolved a sample of ore containing an unknown amount of iron in HCl(*aq*). The resulting aqueous solution, which contained both Fe^{2+}(*aq*) and Fe^{3+}(*aq*), was poured through a Jones reductor column containing Zn(Hg)(*s*) to convert all iron ions into a single oxidation state. The conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*) in the Jones reductor column is shown in Figure 1.

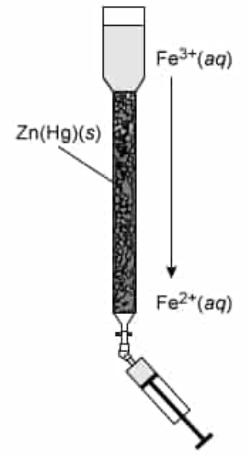

Figure 1 Jones reductor column used for the conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*). The resulting Fe^{2+}(*aq*) solution was collected using a syringe.

To determine the total amount of iron in the sample, the Fe^{2+}(*aq*) solution was then transferred to a container and titrated with 0.1 M $K_2Cr_2O_7$(*aq*), which resulted in the conversion of Fe^{2+}(*aq*) to Fe^{3+}(*aq*) in an irreversible process. A reference electrode was used to monitor the solution potential as a function of $K_2Cr_2O_7$(*aq*) volume added, and the equivalence point of the solution was reached after the addition of 9.50 mL of $K_2Cr_2O_7$(*aq*). A schematic of the experimental setup is shown in Figure 2.

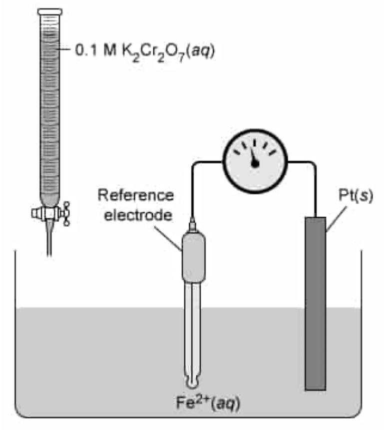

Figure 2 Experimental setup for the titration of Fe^{2+}(*aq*) with $K_2Cr_2O_7$(*aq*) under acidic conditions

In a second experiment, the researcher prepared another sample of Fe^{2+}(*aq*) but used a 0.1 M I_3^-(*aq*) titrant solution instead of $K_2Cr_2O_7$(*aq*).

The standard reduction potentials for the species involved in the experiments are given in Table 1.

Table 1 Standard reduction potentials

Reduction half-reaction	*E*°(V)
$Cr_2O_7^{2-}(aq) + 14\ H^+(aq) + 6\ e^- \rightarrow 2\ Cr^{3+}(aq) + 7\ H_2O(l)$	+1.33
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+0.76
$I_3^-(aq) + 2\ e^- \rightarrow 3\ I^-(aq)$	+0.53
$Zn^{2+}(aq) + Hg(l) + 2\ e^- \rightarrow Zn(Hg)(s)$	−0.76

Which of the following is true regarding spontaneous processes involving I_3^-(*aq*)?

I. I_3^-(*aq*) is a weaker oxidizing agent than $Cr_2O_7^{2-}$(*aq*).

II. I_3^-(*aq*) cannot oxidize Fe^{2+}(*aq*).

III. I_3^-(*aq*) can be reduced in the Jones reductor column.

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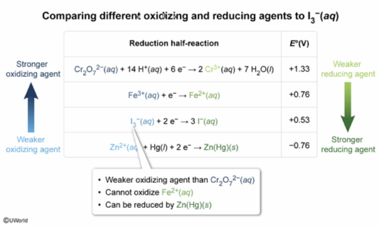
○ A. I and II only [28%]
○ B. I and III only [30%]
○ C. II and III only [12%]
✔ ◉ D. I, II, and III [29%]

Correct
29% answered correctly

Explanation:



During an oxidation–reduction (redox) titration, electrons are lost by one atom (which becomes oxidized) and are gained by another atom (which becomes reduced). The species containing the atom being oxidized is the reducing agent (ie, causes reduction in another atom). Conversely, the species containing the atom being reduced is the **oxidizing agent** (ie, causes oxidation in another atom).

On a table of reduction half-reactions and corresponding reduction potentials ($E°$), the species on the left of the reaction can be reduced and the species on the right of the reaction can be oxidized. The **more positive** the reduction potential, the more likely the reduction reaction will occur. As such, the reduction species has a stronger affinity for electrons and is a **stronger oxidizing agent**. On the other hand, the **less positive** (more negative) the reduction potential is, the more likely the *oxidation* reaction (ie, the reverse of the reduction reaction) will occur. As such, the oxidation species has a stronger tendency to lose electrons and is a **stronger reducing agent**.

According to the reduction potentials in Table 1, $I_3^-(aq)$ is a weaker oxidizing agent than $Cr_2O_7^{2-}(aq)$ because it has a less positive $E°$ than $Cr_2O_7^{2-}(aq)$ (**Number I**). In addition, $I_3^-(aq)$ is a weaker oxidizing agent than $Fe^{3+}(aq)$. As such, $I_3^-(aq)$ cannot oxidize $Fe^{2+}(aq)$ (and therefore the second titration experiment in the passage will not work) because $Fe^{3+}(aq)$ has a stronger affinity for electrons than $I_3^-(aq)$ (**Number II**). On the other hand, $I_3^-(aq)$ is a stronger oxidizing agent than $Zn^{2+}(aq)$ and can oxidize Zn(Hg)(*s*). Therefore, $I_3^-(aq)$ can be reduced in the Jones reductor column (**Number III**).

Educational objective:
A oxidizing agent is a chemical species that gets reduced and causes oxidation in another species in the process. Chemical species with higher, positive standard reduction potentials are more easily reduced and are stronger oxidizing agents than species with lower or negative standard reduction potentials. During a reduction–oxidation reaction, the stronger oxidizing agent gets reduced.

Time spent:0 sec
Subject:
General Chemistry

QID:404101
System:
Solutions and Electrochemistry

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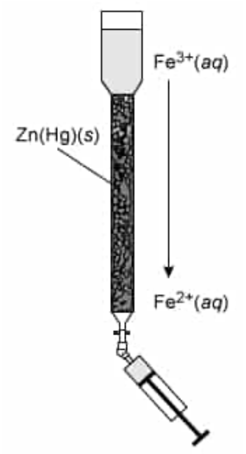

Figure 1 Jones reductor column used for the conversion of Fe^{3+}(*aq*) to Fe^{2+}(*aq*). The resulting Fe^{2+}(*aq*) solution was collected using a syringe.

To determine the total amount of iron in the sample, the Fe^{2+}(*aq*) solution was then transferred to a container and titrated with 0.1 M $K_2Cr_2O_7$(*aq*), which resulted in the conversion of Fe^{2+}(*aq*) to Fe^{3+}(*aq*) in an irreversible process. A reference electrode was used to monitor the solution potential as a function of $K_2Cr_2O_7$(*aq*) volume added, and the equivalence point of the solution was reached after the addition of 9.50 mL of $K_2Cr_2O_7$(*aq*). A schematic of the experimental setup is shown in Figure 2.

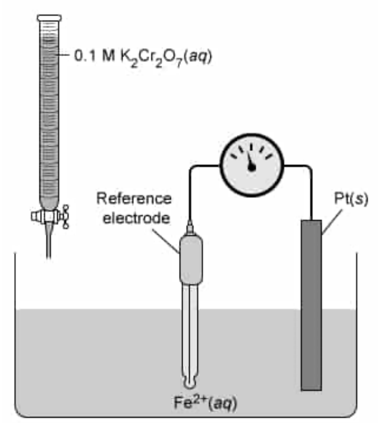

Figure 2 Experimental setup for the titration of Fe^{2+}(*aq*) with $K_2Cr_2O_7$(*aq*) under acidic conditions

In a second experiment, the researcher prepared another sample of Fe^{2+}(*aq*) but used a 0.1 M I_3^-(*aq*) titrant solution instead of $K_2Cr_2O_7$(*aq*).

The standard reduction potentials for the species involved in the experiments are given in Table 1.

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$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+0.76
$I_3^-(aq) + 2\ e^- \rightarrow 3\ I^-(aq)$	+0.53
$Zn^{2+}(aq) + Hg(l) + 2\ e^- \rightarrow Zn(Hg)(s)$	−0.76

Which of the following describes the concentration of Fe^{2+}(*aq*) after 4.75 mL and 9.50 mL of $K_2Cr_2O_7$(*aq*) have been added to the solution, respectively?

- A. $[Fe^{2+}] = [Fe^{3+}]$, $[Fe^{2+}] > [Fe^{3+}]$ [13%]
- ✔ B. $[Fe^{2+}] = [Fe^{3+}]$, $[Fe^{2+}] = 0$ [38%]
- C. $[Fe^{2+}] > [Fe^{3+}]$, $[Fe^{2+}] = 0$ [35%]
- D. $[Fe^{2+}] < [Fe^{3+}]$, $[Fe^{2+}] < [Fe^{3+}]$ [11%]

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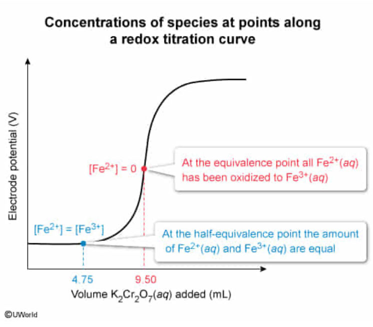
Correct
38% answered correctly

Explanation:



During a typical reduction–oxidation titration, a titrant solution with a known concentration of an oxidizing agent is added to an analyte solution with an unknown concentration to be measured. As the oxidizing agent is added, the analyte becomes oxidized. The **equivalence point** occurs when all the analyte in the solution has been oxidized. This can be detected using an electrode to monitor the solution potential (voltage) or a redox indicator that changes color near the equivalence point. At the **half-equivalence point**, the titrant volume is half the amount required to reach the equivalence point and the molar concentrations of the non-oxidized analyte and oxidized analyte are equal.

The passage states that the equivalence point is reached after 9.50 mL of $K_2Cr_2O_7$(*aq*) (the titrant) has been added to the Fe^{2+}(*aq*) solution (ie, Fe^{2+} is fully oxidized to Fe^{3+}). Consequently, when half this amount is added (ie, 4.75 mL), **$[Fe^{2+}] = [Fe^{3+}]$**, and the molar concentration of Fe^{2+}(*aq*) (ie, $[Fe^{2+}]$) is **zero** after 9.50 mL of $K_2Cr_2O_7$ (*aq*) have been added.

(Choices A, C and D) At the equivalence point (9.50 mL of titrant added), $[Fe^{2+}] = 0$. At the half-equivalence point (4.75 mL of titrant added), $[Fe^{2+}] = [Fe^{3+}]$.

Educational objective:
In a redox titration, the equivalence point occurs when all the analyte in the solution has been oxidized by the added titrant (an oxidizing agent). At the half-equivalence point, the molar concentrations of the non-oxidized analyte and oxidized analyte are equal.

Time spent:0 sec
Subject:
General Chemistry

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System:
Solutions and Electrochemistry

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